

Math 63: Real Analysis

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Problem 1. Prove that a differentiable real-valued function on $\mathbb{R}$ with bounded derivative is uniformly continuous.

Problem 2. Let f be a real-valued function on an open subset U of $\mathbb{R}$ that is twice differentiable at $x_0 \in U$. Show that if $f'(x_0) = 0$ and $f''(x_0) < 0$ ($f''(x_0) > 0$) then the restriction of f to some open ball of center x_0 attains a maximum (minimum) at x_0 .

Problem 3. A real-valued function on an open interval in $\mathbb{R}$ is called convex if no point on the line segment between any two points of its graph lies below the graph. If the function is differentiable, this condition is known to be equivalent to the condition that no point of the graph lie below any point of any tangent to the graph. If the function is twice differentiable the condition is known to be equivalent to the second derivative of the function being nonnegative at all points. State these conditions in precise analytic terms and prove them.

Problem 4. Prove that $\int_0^1 f(x)dx = 0$ if $f(1/n) = 1$ for $n = 1, 2, 3, \dots$ and $f(x) = 0$ for all other x .

Problem 5. Does $\int_0^1 f(x)dx$ exist for the following function f :

$$f(x) = \begin{cases} 0 & \text{if } x \notin \mathbb{Q} \\ \frac{1}{q} & \text{if } x = \frac{p}{q} \in \mathbb{Q} \text{ where } \gcd(p, q) = 1 \end{cases}$$